

SHE Culture
ISSN: 2736-0350

Providing insights to innovation in research and reliable industry problem solving

The SHE Culture Journal (ISSN 2736-0350) is the key communication and knowledge exchange tool of the "EMT International Knowledge Exchange Forum for Leadership & Behavioral Safety Dissemination" which serves as a knowledge sharing and EHS dissemination media set up in honour of Professor E. Scott Geller (Distinguished Alumni Professor, Center for Applied Behavior Systems, Virginia Tech, Blacksburg, United States of America (USA), Founder, Actively Caring for People (AC4P) & founding partner, Safety Performance Solutions, SPS, USA

A Publication sponsored by the EMT - CREED, International Projects Team
EMT Centre for Research, Environment & Development, Inc.,
Registered & incorporated in the State of Texas, USA
File Number: 806707886
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EMT-SCUT
The Chronic Unease & Safety Culture Team

Prof. Patricks-E, Chinemerem
Editorial Board Advisor

El Chumo Inc
Andover Park Drive, Houston, Texas, USA

IFTDO @ AM 630 Member
B7 Kestrel Court, Harbour Road, Portishead, BS20 7AN, UK
S-50, Greater Kailash I, New Delhi 110048, India
www.iftdo.org

Professor E. Scott Geller
Alumni Distinguished Professor
Center for Applied Behavior Systems
Virginia Tech University
Blacksburg, VA, USA.

Professor Ludwig Timothy
Appalachian State University,
United States of America
Safety-Doc.com

Catherine C. Patricks
Website Content Administrator

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Professor Ugwoha Ejikeme
International Review Consultant, Nigeria
Centre for Occupational Health, Safety & Environment- COHSE
University of Port Harcourt, Nigeria

Correspondence
emtsheculturejournal@gmail.com
environmenttargets@yahoo.com
www.environmenttargets.com
www.iftdo.org

Published by:
EMT Centre for Research, Environment Education & Development, Inc.
in collaboration with
Environment & Millennium Targets Ltd

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This publication has been edited and reviewed by the SHE Culture Journal (ISSN 2736-0350) editorial and technical review team in consultation with the "EMT Prof. E. Scott Geller International Knowledge Exchange Forum for Safety Leadership & Behavioral Safety Dissemination" ***a knowledge sharing and EHS dissemination media set up in honour of Professor E. Scott Geller (Distinguished Alumni Professor, Department of Psychology, Virginia Tech, Blacksburg, United States of America (USA), Founder & Partner, Safety Performance Solutions, SPS, USA***



Celebrating the academic and consulting giant of Behavioural & Psychological Safety after 56 years on Faculty
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Professor E. Scott Geller (Distinguished Alumni Professor, Department of Psychology, Virginia Tech, Blacksburg, retired in 2025 after 56 years of championing psychological and behavioral safety and contributing to the growth of knowledge and body of literature in this ever-evolving field of learning and academics



GellerFest (May, 2025): Celebrating the distinguished Alumni Professor E. Scott Geller after 56 years at Virginia Tech

CORRELATION AND VARIATION ANALYSIS IN GREENHOUSE GAS EMISSION SOURCES AND REDUCTION POTENTIALS AMONG EMITTING FACILITIES IN PORT HARCOURT, NIGERIA

[1]Ngaju Chidinma L., [2]Ugwoha Ejikeme, & [3]Patricks Chinemerem
Centre for Occupational Health, Safety and Environment (COHSE), University of Port Harcourt, Rivers State, Nigeria.

Editor's Note & Commentary: The authors provided useful information on the strength of association and variations in Greenhouse gas (GHG) sources and opportunities to mitigate (by reducing GHG emissions) and adapting to the impacts of climate change. The identification of activities that emit GHG's within an organization's operation and the proposal of actions to reduce same GHG emission are important components of a climate change risk assessment.

Disclaimer: The data and facts presented here were not subjected to any verification by the SHE Culture journal team and as such are purely research-based findings by the authors within their selected study area.

The Authors:
[1]NGAJU, CHIDINMA.L., is the lead and corresponding author. She is a PhD student of the Center for Occupational Health, Safety and Environment (COHSE), University of Port Harcourt, Choba, Nigeria.
E-Mail: ngajuchidinmaloveth@gmail.com, Tel: +234 -70 63876289

[2]UGWOHA. EJIKEME., PhD., is a Professor and the Assistant Director at the Centre for Occupational Health, Safety and Environment (COHSE), University of Port Harcourt, Choba, Nigeria.
E-Mail: ugwohaej@gmail.com , Tel: +234-80 31181050

[3]PATRICKS E.C., PhD., is a professor of Loss Prevention & Human Factors Engineering. He is a member of the academic board and visiting lecturer at the Center for Occupational Health, Safety and Environment, University of Port Harcourt, Choba, Nigeria. He is an industry consultant for safety leadership, human performance, EHS Research & knowledge exchange and a member of EMT/IFTDO International projects committee representing IFTDO in the TC/260, WG-11- ISO 30422:2022 new standard for Learning & Development. He is also a visiting professor to the World Bank Africa Centre of Excellence in Oilfield Chemicals Research (ACE-CEFOR), University of Port Harcourt, Nigeria.
E-Mail: chinemerem.patricks@yahoo.com, Tel: +234- 8038691606

Abstract
Greenhouse Gas emissions from industrial activities continue to present significant climate change induced challenges in vulnerable regions. This study assessed major greenhouse gas emission sources, mitigation opportunities across selected emitting sectors in Port Harcourt, Nigeria. A descriptive and correlational research design was adopted. Purposive sampling was adopted in the selection and sampling of twenty-five major greenhouse gas-emitting facilities. Primary data were collected through structured questionnaires and focus group interviews. Descriptive statistics and correlation analysis were employed to analyze the data at a significance level of 0.05. The facilities exhibited broadly comparable categories of GHG generating activities, types and mitigation options, reflecting their common sectoral characteristics but despite this similarity in the categories identified, the strength of association in their quantitative emission reduction profiles varied, indicating that the relative magnitude of emissions and reduction potential was not uniform across the sampled facilities. Significant negative correlation was observed between source and GHG-type scores across all three manufacturing facilities (r = -.233 to -.338, p < .01), while mitigation scores differed significantly among facilities, F (2, 400) = 19.51, p < .001. Oil and gas facilities showed no significant relationship among the study constructs, although GHG-source scores differed significantly, F (2, 404) = 5.62, p = .004. Transportation facilities demonstrated comparatively consistent response patterns, with no significant difference between facilities. Solid waste exhibited the greatest heterogeneity, with multiple positive correlations and significant facility differences in GHG types, F (2, 395) = 11.38, p < .001; health and safety impacts, F (2, 395) = 3.54, p = .030; and mitigation options, F (2, 395) = 3.32, p = .037. The findings demonstrate that relationships between identified GHG characteristics and mitigation opportunities are facility- and sector-specific rather than uniformly strong across the study population. Consequently, facility-specific assessments are recommended to complement sector-wide GHG mitigation strategies.

Keywords: greenhouse gas, emission sources, mitigation opportunities, reduction potential, decarbonization, industrial facilities, net-zero emissions.

Citation: This paper may be cited as:
Ngaju, C.L., Ugwoha, E., & Patricks, E.C. (2026). Correlation and variation analysis in greenhouse gas emission sources and reduction potentials among emitting facilities in Port Harcourt, Nigeria. *SHE Culture Journal*, 8(5), 9-23

Introduction

Climate change represents one of the most significant environmental, economic, and social challenges of the twenty-first century. Rising concentrations of greenhouse gases (GHGs), particularly carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O), have accelerated global warming and contributed to increasingly severe climate impacts, including rising temperatures, altered precipitation patterns, biodiversity loss, sea-level rise, and frequent extreme weather events (Intergovernmental Panel on Climate Change [IPCC], 2023).

It is a record fact that the earth's climate system has greatly changed on both global and regional scales since the pre-industrial era, with some of these changes linked and attributed to human activities (Patricks, 2020). Several other scientific evidence indicates that anthropogenic activities remain the dominant drivers of these atmospheric changes, with energy production and industrial activities accounting for a substantial proportion of global emissions.

According to (Patricks, 2025), the extent and magnitude of climate change impacts are larger for each additional fraction of warming than estimated in previous assessments; so are the risks projected for the future. We risk crossing what we call tipping points where we approach the limits of what species and ecosystems can tolerate. Therefore, it is important that robust and ambitious reductions in GHG emissions are carried out to avoid irreversible changes in the climate system.

Climate change is driven primarily by the accumulation of greenhouse gases resulting from anthropogenic activities, including energy production and consumption, industrial processes, transportation, agriculture, land-use change, and waste management. The Intergovernmental Panel on Climate Change (IPCC) recognizes substantial mitigation opportunities across these sectors but emphasizes that appropriate mitigation pathways vary according to technologies, operational systems, infrastructure, and socioeconomic conditions (IPCC, 2022, 2023).

Industrial activities are particularly important because their emissions arise from several distinct pathways. According to the IPCC (2022), industrial GHG emissions can originate from fuel combustion, industrial processes, product use, and waste. Energy efficiency, material efficiency, circular economy approaches, low-carbon energy sources, process changes, carbon capture and storage, and other technological interventions constitute potential pathways for industrial decarbonization (IPCC, 2022).

A scientifically important distinction must therefore be maintained between GHG emission sources and the gases generated by those sources. Activities such as fuel combustion, gas flaring, venting, transportation, industrial processing, and waste decomposition constitute emission-generating sources or processes. Carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), and fluorinated gases, by contrast, constitute GHG types. Treating a GHG such as methane and an activity such as gas flaring as equivalent "sources" would therefore conflate two conceptually different elements of an emission pathway.

The issue is especially important in Nigeria because the country's emission profile encompasses several economic sectors and emission-generating activities. Nigeria's national GHG reporting recognizes energy, industrial processes and product use, agriculture, forestry and other land use, and waste as important components of the national emissions profile (Federal Republic of Nigeria, 2023). Nigeria's current climate commitments similarly distinguish mitigation measures relating to energy-related fuel combustion, fugitive emissions, industrial processes, transportation, and waste management (Federal Republic of Nigeria, 2025).

These challenges are relevant to Port Harcourt, a major industrial and commercial centre in the Niger Delta. Recent research has demonstrated considerable spatial variability in environmental pollution across Port Harcourt, influenced by industrial operations, transportation activity, energy use, and other anthropogenic processes (Eyetan et al., 2025). Although conventional air pollutants should not be interpreted as equivalent to GHGs, these findings demonstrate the importance of recognizing spatial and operational heterogeneity when investigating environmental emissions.

Energy-related research conducted in Port Harcourt had similarly identified substantial reliance on fossil-fuel-based energy systems, particularly generator use, while emphasizing renewable energy, grid improvements, and cleaner energy alternatives as important components of transition toward net-zero emissions (Adiama et al., 2025). In the transportation sector, Akujor et al. (2022) observed significant dependence on fossil fuels in Nigeria and highlighted mass transit, low-carbon fuels, improved rail and water transport, and electric vehicles as potential decarbonization pathways.

The oil and gas sector presents a different operational profile. Gas flaring, for example, remains an important emission-generating activity within Nigeria's petroleum sector and has attracted policy attention because reducing routine flaring can contribute to national climate mitigation objectives (Afinotan, 2022). Similarly, methane emissions associated with oil and gas extraction may arise through activities such as venting, fugitive releases, and incomplete combustion, highlighting the need to distinguish the gas emitted from the activity responsible for its release (Raimi et al., 2025).

Waste management presents another distinct mitigation challenge. Waste decomposition, open dumping, uncontrolled burning, collection, and transportation may produce different emission profiles, whereas improved recovery, recycling, composting, engineered disposal systems, and other sustainable waste-management approaches offer potential mitigation pathways. Recent Nigerian research has demonstrated that alternative waste-management scenarios can produce substantially different GHG outcomes, reinforcing the importance of facility- and technology-specific mitigation planning (Lawal et al., 2026).

Consequently, facilities operating within the same economic sector cannot automatically be assumed to possess identical emission-source profiles or equivalent mitigation opportunities. Operational scale, technology, energy sources, waste-management systems, management practices, and existing environmental controls can vary considerably from one facility to another.

This creates an important empirical question: Do facilities operating within the same sector demonstrate similar relationships among their identified GHG types and sources, and do they possess comparable opportunities for emission reduction?

Addressing this question is necessary because a strong sector-wide mitigation recommendation may be inappropriate where substantial facility-level differences exist. Conversely, where facilities demonstrate statistically comparable profiles, common sector-level interventions may be justified.

The present study therefore investigated facility-level relationships and variations among manufacturing, oil and gas, transportation, and solid waste facilities in Port Harcourt, with particular attention to identified GHG emission sources and mitigation opportunities.

The objectives of this study are to:

1. Identify the major greenhouse gas emission sources, GHG types and mitigation opportunities among selected facilities in the manufacturing, oil and gas, transportation and solid waste sectors in Port Harcourt.

2. Examine the relationships among identified greenhouse gas emission sources, GHG types and mitigation opportunities within facilities in each sector.

3. Determine whether significant variations exist in identified GHG emission sources and mitigation opportunities among facilities within each sector.

Methodology

The study adopted a cross-sectional quantitative survey design to investigate GHG emission characteristics and potential mitigation opportunities among selected emitting facilities in Port Harcourt, Rivers State, Nigeria. Facilities were drawn from four sectors: manufacturing, oil and gas, transportation, and solid waste management.

Three facilities were evaluated within each sector, designated Facilities A, B and C, resulting in a total of 12 facilities. Respondent-level data were analyzed separately according to facility and sector.

2.2 Data Collection

Primary data were obtained through a structured questionnaire administered to respondents within the sampled facilities. The questionnaire captured information under four broad constructs:

- identified sources of GHG emissions;
- types of GHG associated with facility operations;
- perceived health and safety impacts; and
- identified GHG mitigation options or reduction opportunities.

Responses were measured using a five-point Likert scale. Composite respondent-level scores were derived from questionnaire items belonging to the respective constructs.

Facility sample sizes varied slightly across sectors. Manufacturing comprised 403 respondents distributed across Facilities A (n = 134), B (n = 134), and C (n = 135). Oil and gas comprised 407 respondents across Facilities A (n = 135), B (n = 135), and C (n = 137). Transportation comprised 401 respondents across Facilities A (n = 133), B (n = 133), and C (n = 135), while solid waste comprised 398 respondents across Facilities A (n = 132), B (n = 132), and C (n = 134).

2.3 Statistical Analysis

Data were analyzed using descriptive and inferential statistical techniques. Means and standard deviations were used to describe composite scores within facilities.

Pearson product-moment correlation was used to investigate relationships among GHG emission-source scores, identified GHG types, health and safety impacts, and mitigation-option scores separately within each facility.

A one-way analysis of variance (ANOVA) was used to determine whether mean scores differed significantly among Facilities A, B and C within each sector. Where the omnibus ANOVA was significant, Tukey's Honestly Significant Difference (HSD) test was applied to determine the specific facility pairs responsible for the observed differences.

Eta-squared (η^2) was reported to quantify the magnitude of between-facility effects. Chi-square tests of independence and Cramer's V were additionally used as complementary analyses to assess differences in categorical Likert-response distributions among facilities. Statistical significance was established at $p < .05$.

Results

Manufacturing Sector

Pearson correlation analysis revealed a consistent statistically significant negative correlation between identified GHG emission sources and GHG-type scores in each manufacturing facility. The relationships were r = -.338 ($p < .001$) for Facility A, r = -.301 ($p < .001$) for Facility B, and r = -.233 (p = .006) for Facility C. Relationships involving sources and mitigation options were not significant in Facilities A (r = .008, p = .925), B (r = .016, p = .853), or C (r = -.084, p = .332). Other correlations involving health and safety impacts and mitigation options were similarly non-significant.

ANOVA demonstrated that the three manufacturing facilities were statistically comparable for GHG sources, F (2, 400) = .076, p = .927; GHG types, F (2, 400) = .024, p = .977; and health and safety impacts, F (2, 400) = 1.488, p = .227. However, mitigation options differed significantly, F (2, 400) = 19.512, $p < .001$, η^2 = .089. Tukey HSD showed that Facilities A (M = 4.067) and B (M = 4.063) both recorded significantly higher mitigation scores than Facility C (M = 3.648), while A and B did not differ significantly.

Thus, the manufacturing facilities displayed broadly similar GHG characteristics, but their perceived mitigation opportunities were not uniform.

Table 1. Descriptive Statistics

Variable	Facility	N	Mean	Std. Deviation
GHG Sources	A	134	3.595	.725
	B	134	3.629	.738
	C	135	3.615	.746
GHG Types	A	134	3.418	.819
	B	134	3.403	.827
	C	135	3.424	.816
HSI	A	134	3.694	.695
	B	134	3.778	.635
	C	135	3.637	.690
Mitigation Options	A	134	4.067	.610
	B	134	4.063	.570
	C	135	3.648	.709

As displayed in Table 2: The Pearson correlation analysis showed significant negative correlation between GHG Sources and GHG Types in Facility A, r (132) = -.338, p < .001; Facility B, r (132) = -.301, p < .001; and Facility C, r (133) = -.233, p = .006. No other relationships were statistically significant.

Table 2. Correlation

Variables	Facility	Pearson Correlation (r)	Sig. (2-tailed)	N
Sources × Types	A	-.338**	<.001	134
	B	-.301**	<.001	134
	C	-.233**	.006	135
Sources × HSI	A	-.023	.792	134
	B	.107	.220	134
	C	-.001	.995	135
Sources × Mitigation	A	.008	.925	134
	B	.016	.853	134
	C	-.084	.332	135
Types × HSI	A	.037	.669	134
	B	-.006	.949	134
	C	.094	.278	135
Types × Mitigation	A	-.025	.778	134
	B	-.103	.239	134
	C	.124	.151	135
HSI × Mitigation	A	.053	.541	134
	B	-.002	.979	134
	C	-.080	.357	135

** Correlation is significant at the .01 level (2-tailed).

Table 3. ANOVA

Variable	df Between	df Within	F	Sig.	η^2
Sources	2	400	.076	.927	.000
Types	2	400	.024	.977	.000
HSI	2	400	1.488	.227	.007
Mitigation Options	2	400	19.512	<.001	.089

The only significant facility difference was for Mitigation Options, F (2, 400) = 19.512, p < .001, η^2 = .089.

Table 4. Tukey HSD Multiple Comparisons: Mitigation

(I) Facility	(J) Facility	Mean Difference (I-J)	Sig.	95% CI Lower	95% CI Upper
A	B	.004	.999	-.178	.186
A	C	.419	<.001	.238	.601
B	C	.415	<.001	.234	.597

Facilities A and B did not differ, whereas both recorded significantly higher mitigation scores than Facility C.

Oil and Gas Sector

Unlike manufacturing, none of the correlations among GHG sources, GHG types, health and safety impacts, and mitigation options was statistically significant within the three oil and gas facilities. For example, the correlation between identified emission sources and mitigation options was r = .040 (p = .643) for Facility A, r = .041 (p = .640) for Facility B, and r = .002 (p = .980) for Facility C.

The ANOVA nevertheless identified a significant difference between facilities in GHG-source scores, F (2, 404) = 5.620, p = .004, η^2 = .027. Facility A had the highest source score (M = 3.730), compared with Facility B (M = 3.412) and Facility C (M = 3.568). Tukey HSD confirmed that the significant difference occurred between Facilities A and B (mean difference = .319, p = .003). No significant differences were observed for GHG types, health and safety impacts, or mitigation options

This finding suggests that emission-source characteristics varied to some extent among oil and gas facilities, although their mitigation-option scores were comparatively consistent.

Table 5. Correlations

Variables	Facility A r (Sig.)	Facility B r (Sig.)	Facility C r (Sig.)
Sources × Types	-.073 (.402)	-.030 (.733)	-.049 (.568)
Sources × HSI	.065 (.454)	.039 (.656)	-.159 (.063)
Sources × Mitigation	.040 (.643)	.041 (.640)	.002 (.980)
Types × HSI	-.151 (.080)	-.089 (.306)	-.081 (.345)
Types × Mitigation	-.091 (.295)	.092 (.290)	.152 (.077)
HSI × Mitigation	.127 (.142)	.118 (.175)	.162 (.059)

None of the relationships was statistically significant at $p < .05$.

Table 6. Descriptive Statistics and ANOVA

Variable	Facility A M ± SD	Facility B M ± SD	Facility C M ± SD	F (2,404)	Sig.	η^2
Sources	3.730 ± .686	3.412 ± .853	3.568 ± .793	5.620	.004	.027
Types	3.076 ± .726	3.012 ± .691	3.111 ± .714	.678	.508	.003
HSI	3.534 ± .616	3.487 ± .606	3.503 ± .642	.205	.814	.001
Mitigation	3.611 ± .689	3.613 ± .726	3.631 ± .779	.032	.969	.000

Only GHG Sources showed a statistically significant difference among facilities.

Table 7. Tukey HSD: GHG Sources

(I) Facility	(J) Facility	Mean Difference	Sig.	95% CI
A	B	.319	.003	.095 to .542
A	C	.163	.200	-.060 to .385
B	C	-.156	.227	-.379 to .067

Thus, Facility A recorded significantly higher GHG-source scores than Facility B.

Transportation Sector

Transportation demonstrated the greatest consistency among facilities. Correlations were predominantly non-significant. The only statistically significant relationship was observed in Facility C between health and safety impacts and mitigation options, $r = .225$, $p = .009$, representing a weak positive association.

One-way ANOVA confirmed the relative uniformity of the three transportation facilities. No significant differences were observed in identified GHG sources, F (2, 398) = .074, $p = .928$; GHG types, F (2, 398) = .234, $p = .792$; health and safety impacts, F (2, 398) = .180, $p = .836$; or mitigation options, F (2, 398) = .868, $p = .420$.

The results therefore suggest comparatively homogeneous GHG-related response patterns and mitigation perceptions across the sampled transportation facilities.

Table 8. Correlations

Variables	Facility A r (Sig.)	Facility B r (Sig.)	Facility C r (Sig.)
Sources × Types	-.078 (.369)	-.031 (.725)	-.006 (.950)
Sources × HSI	-.063 (.468)	.079 (.367)	-.049 (.570)
Sources × Mitigation	-.161 (.064)	.081 (.351)	-.164 (.057)
Types × HSI	.071 (.415)	-.147 (.091)	-.085 (.326)
Types × Mitigation	.083 (.341)	-.080 (.358)	-.141 (.104)
HSI × Mitigation	.120 (.171)	.157 (.071)	.225 (.009)

Only Facility C showed a significant correlation between HSI and Mitigation, $r = .225$, $p = .009$.

Table 9. Descriptive Statistics and ANOVA

Variable	Facility A M ± SD	Facility B M ± SD	Facility C M ± SD	F (2,398)	Sig.	η^2
Sources	3.624 ± .717	3.612 ± .728	3.590 ± .741	.074	.928	.000
Types	3.382 ± .831	3.412 ± .837	3.450 ± .797	.234	.792	.001
HSI	3.709 ± .692	3.744 ± .643	3.756 ± .670	.180	.836	.001
Mitigation	4.064 ± .617	4.071 ± .567	3.985 ± .598	.868	.420	.004

No statistically significant facility differences occurred. Therefore, interpreting a Tukey HSD post-hoc comparison for this sector is not justified.

Solid Waste Sector

Solid waste demonstrated the greatest number of statistically significant relationships. In Facility A, health and safety impacts were positively related to mitigation options, $r = .319$, $p < .001$. In Facility B, significant positive relationships existed between sources and GHG types ($r = .271$, $p = .002$), GHG types and health and safety impacts ($r = .243$, $p = .005$), and GHG types and mitigation options ($r = .332$, $p < .001$). Facility C similarly demonstrated significant relationships between sources and GHG types ($r = .301$, $p < .001$) and between GHG types and health and safety impacts ($r = .248$, $p = .004$).

ANOVA revealed no significant difference among solid waste facilities for GHG-source composite scores, F (2, 395) = .509, $p = .601$. Significant differences were, however, observed for GHG types, F (2, 395) = 11.375, $p < .001$, $\eta^2 = .055$; health and safety impacts, F (2, 395) = 3.536, $p = .030$, $\eta^2 = .018$; and mitigation options, F (2, 395) = 3.323, $p = .037$, $\eta^2 = .017$.

Tukey HSD demonstrated that Facility A recorded significantly higher GHG-type scores than Facilities B and C. For health and safety impacts and mitigation options, significant differences occurred specifically between Facilities A and B.

Table 10. Correlations

Variables	Facility A r (Sig.)	Facility B r (Sig.)	Facility C r (Sig.)
Sources × Types	.057 (.517)	.271 (.002)	.301 (<.001)
Sources × HSI	.034 (.698)	.067 (.446)	.113 (.193)
Sources × Mitigation	-.146 (.094)	-.006 (.944)	.012 (.892)
Types × HSI	.121 (.168)	.243 (.005)	.248 (.004)
Types × Mitigation	-.031 (.725)	.332 (<.001)	.017 (.842)
HSI × Mitigation	.319 (<.001)	.100 (.254)	.084 (.337)

The Solid Waste sector showed the greatest number of significant correlations.

Table 11. Descriptive Statistics and ANOVA

Variable	Facility A M ± SD	Facility B M ± SD	Facility C M ± SD	F (2,395)	Sig.	η^2
Sources	3.717 ± .806	3.806 ± .873	3.813 ± .904	.509	.601	.003
Types	3.286 ± .834	2.877 ± .750	2.890 ± .793	11.375	<.001	.055
HSI	3.644 ± .707	3.430 ± .713	3.457 ± .718	3.536	.030	.018
Mitigation	3.835 ± .711	3.621 ± .669	3.672 ± .735	3.323	.037	.017

Table 12. Tukey HSD Multiple Comparisons

Variable	Comparison	Mean Difference	Sig.	95% CI
GHG Types	A vs B	.409	<.001	.180 to .639
GHG Types	A vs C	.396	<.001	.167 to .625
GHG Types	B vs C	-.013	.990	-.242 to .216
HSI	A vs B	.214	.040	.008 to .420
HSI	A vs C	.187	.083	-.019 to .392
HSI	B vs C	-.027	.948	-.233 to .178
Mitigation	A vs B	.214	.038	.010 to .418
Mitigation	A vs C	.164	.143	-.040 to .367
Mitigation	B vs C	-.050	.829	-.254 to .153

Table 13. Pearson Chi-Square Tests of Facility × Response Category

Sector	Construct	Pearson χ^2	df	Asymp. Sig. (2-sided)	Cramer's V
Manufacturing	Sources	2.547	8	.959	.032
Manufacturing	Types	1.059	8	.998	.018
Manufacturing	HSI	10.574	8	.227	.057
Manufacturing	Mitigation	43.983	8	<.001	.117
Oil & Gas	Sources	85.838	8	<.001	.145
Oil & Gas	Types	4.980	8	.760	.035
Oil & Gas	HSI	2.379	8	.967	.020
Oil & Gas	Mitigation	2.056	8	.979	.025
Transportation	Sources	2.175	8	.975	.030
Transportation	Types	1.541	8	.992	.022
Transportation	HSI	1.195	8	.997	.019
Transportation	Mitigation	5.497	8	.703	.041
Solid Waste	Sources	35.605	8	<.001	.122
Solid Waste	Types	82.745	8	<.001	.161
Solid Waste	HSI	57.616	8	<.001	.135
Solid Waste	Mitigation	58.292	8	<.001	.135

All expected frequencies exceeded 5 in these construct-level contingency tables.

Chi-square tests of independence were conducted to determine whether respondents' ratings of GHG emission sources, GHG types, health and safety impacts, and mitigation options were associated with facility membership within each sector. In the Manufacturing sector, a significant association was observed only for mitigation options, χ^2 (8) = 43.98, p < .001, Cramer's V = .117, whereas GHG sources, types, and health and safety impacts showed no significant facility-level associations. Within the Oil and Gas sector, a significant association was found for GHG emission sources, χ^2 (8) = 85.84, p < .001, Cramer's V = .145, while the remaining constructs were not significant. No significant associations were observed for any of the four constructs within the Transportation sector, indicating comparable response distributions across its three facilities. In contrast, the Solid Waste sector demonstrated significant facility-level associations for GHG sources, χ^2 (8) = 35.61, p < .001, Cramer's V = .122; GHG types, χ^2 (8) = 82.74, p < .001, Cramer's V = .161; health and safety impacts, χ^2 (8) = 57.62, p < .001, Cramer's V = .135; and mitigation options, χ^2 (8) = 58.29, p < .001, Cramer's V = .135. Overall, the results demonstrate that facility-level response patterns are largely consistent within Transportation, selectively different within Manufacturing and Oil and Gas, and most heterogeneous within the Solid Waste sector.

Table 14. Consolidated Inferential Findings

Sector	Correlation Finding	Significant ANOVA Result	Main Facility Difference
Manufacturing	Sources × Types significant and negative in A, B and C	Mitigation: F (2,400) = 19.512, p < .001, η^2 = .089	A > C; B > C
Oil & Gas	No significant correlations	Sources: F (2,404) = 5.620, p = .004, η^2 = .027	A > B
Transportation	HSI × Mitigation significant only in C, r = .225, p = .009	None	Facilities statistically comparable
Solid Waste	Multiple positive significant correlations	Types : p < .001 ; HSI : p = .030 ; Mitigation : p = .037	Facility A principally differentiated from B/C

Discussion

Major greenhouse gas emission sources, GHG types and mitigation opportunities among selected facilities in the selected sectors
The descriptive variation supports the proposition that GHG characteristics are partly dependent on facility-specific operational conditions. This is consistent with IPCC (2022), which identifies multiple emission pathways within industry, including fuel combustion, process emissions, product use, and waste, while also emphasizing that viable mitigation options depend on the processes and technologies involved.

The findings are also consistent with Nigerian evidence demonstrating that emission profiles can vary considerably according to energy source and operational scale. Research in Port Harcourt, for example, has identified distinctions in energy consumption and GHG profiles among different categories of energy users, supporting differentiated rather than uniform mitigation interventions (Okon, 2025).

Similarly, Nigerian transportation depends heavily on fossil fuels, while proposed decarbonization strategies include electric mobility, low-carbon fuels, public transportation, rail, and waterways (Akujor et al., 2022). This indicates that emission-producing activities and their corresponding mitigation opportunities differ according to sectoral and operational context.
The Solid Waste results are especially consistent with the proposition that waste-management systems are heterogeneous. Waste transportation, collection efficiency, disposal methods, recycling, resource recovery, and landfill configuration can differ substantially even within one metropolitan waste-management system (Ogboeli et al., 2025). This provides a reasonable contextual explanation for the relatively greater facility variation observed within Solid Waste.

Relationships Among GHG Sources, Types, and Mitigation Opportunities Within Facilities

Manufacturing demonstrated significant negative relationships between identified GHG-source and GHG-type composite scores in Facility A, r (132) = -.338, $p < .001$; Facility B, r (132) = -.301, $p < .001$; and Facility C, r (133) = -.233, p = .006. No significant relationships were observed between source and mitigation scores. The consistency of the negative coefficients suggests that respondents giving relatively higher ratings to the emission-source items tended to provide somewhat lower ratings to GHG-type items.

This finding differs from what might be expected if the variables represented directly measured physical emissions. The IPCC (2022) clearly identifies fuel combustion, industrial processes, product use, and waste as activities through which industrial GHG emissions arise. Consequently, the negative relationships observed here are more appropriately interpreted as differences in respondents' recognition or assessment of sources and GHG types than as evidence of an inverse physical emissions relationship.

No statistically significant correlations were identified among GHG sources, GHG types, health and safety impacts, and mitigation opportunities across the three Oil and Gas facilities. For source and mitigation scores, correlations ranged from r = .002 to .041, with all p values exceeding .05. This indicates that greater recognition of emission-generating activities did not necessarily correspond with greater recognition of mitigation opportunities within the surveyed Oil and Gas facilities.

This finding highlights an important practical issue. Oil and gas operations can involve multiple emission pathways requiring different mitigation technologies. Measures aimed at gas flaring, for instance, may differ from those addressing fugitive methane releases, combustion efficiency, or operational energy use.

Afinotan (2022) notes the particular importance of gas-flaring regulation to Nigeria's climate mitigation efforts, while Raimi et al. (2025) emphasize the importance of methane monitoring and management within Nigerian oil and gas activities. Therefore, the absence of a general correlation between source and mitigation scores may reflect the fact that mitigation potential is technology- and activity-specific rather than simply proportional to the recognition of emission sources.

Most relationships within Transportation were non-significant. The sole significant relationship was observed in Facility C between health and safety impacts and mitigation options, r = .225, p = .009. Although statistically significant, r = .225 indicates only a weak positive relationship. It therefore suggests that respondents demonstrating greater recognition of health and safety impacts tended to report somewhat greater recognition of mitigation opportunities, but the constructs were not strongly related.

The general absence of strong relationships is consistent with the complexity of transport-sector decarbonization. Akujor et al. (2022) reported extensive fossil-fuel dependence in Nigeria's transport sector and identified multiple potential mitigation pathways rather than a single intervention, including mass transit, rail development, electric vehicles, low-carbon fuel systems, and improved transport infrastructure.

Solid Waste demonstrated the most extensive network of statistically significant positive relationships. In Facility A, health and safety impacts were positively associated with mitigation options, r = .319, $p < .001$. Facility B demonstrated significant positive relationships for Sources × Types (r = .271, p = .002), Types × HSI (r = .243, p = .005), and Types × Mitigation (r = .332, $p < .001$). Facility C recorded significant positive relationships for Sources × Types (r = .301, $p < .001$) and Types × HSI (r = .248, p = .004).

These results suggest that respondents' recognition of GHG-related issues is comparatively interconnected within the Solid Waste sector. For example, within Facility B, greater recognition of GHG types was associated with greater recognition of mitigation opportunities.

This is broadly consistent with empirical research showing that the emissions from waste dump sites and landfills are closely connected to disposal and recovery pathways. Lawal et al. (2026) demonstrated substantial differences in GHG outcomes among open dumping, open burning, landfilling, composting, recycling, anaerobic digestion, and alternative waste-management scenarios in Nigeria. Similarly, operational research in Port Harcourt has identified substantial variation in waste collection and transportation efficiency and has emphasized route optimization, fleet management, public-private partnerships, and improved service organization as important components of sustainable waste management (Ogboeli et al., 2025). The present results therefore support a facility-specific interpretation: the relationships among emission sources, GHG types, impacts, and mitigation opportunities appear more pronounced in Solid Waste than in the other sectors investigated.

Variation Among Facilities Within Each Sector

In the manufacturing sector, there were no significant facility-level differences for GHG sources, F (2, 400) = .076, p = .927; GHG types, F (2, 400) = .024, p = .977; or health and safety impacts, F (2, 400) = 1.488, p = .227. Mitigation opportunities, however, differed significantly, F (2, 400) = 19.512, p < .001, η^2 = .089. Tukey HSD indicated that Facilities A and B had significantly higher mitigation scores than Facility C, while Facilities A and B did not differ significantly. The effect size η^2 = .089 shows that facility components accounted for a meaningful, although not dominant, proportion of variation in with respect to mitigation within the Manufacturing sector. This aligns with the IPCC's conclusion that the ability to decarbonize industrial production depends on factors such as process configuration, energy efficiency, infrastructure, technology availability, material efficiency, low-carbon energy, and carbon-management technologies (IPCC, 2022). Consequently, manufacturing facilities operating under broadly similar sectoral conditions may nevertheless differ substantially in their readiness or opportunities for GHG reduction and climate change mitigation.

In the oil and gas sector, a statistically significant facility difference was observed for GHG emission-source scores, F (2, 404) = 5.620, p = .004, η^2 = .027. Facility A (M = 3.730) scored significantly higher than Facility B (M = 3.412), while Facility C did not differ significantly from either. Differences were not significant for GHG types, HSI, or mitigation opportunities. This finding is consistent with the operational diversity characteristic of oil and gas systems. Differences in production activities, flaring practices, venting, combustion equipment, fugitive-emission management, and facility infrastructure may generate different source profiles even where facilities operate within the same broad sector. Nigeria's climate policy specifically distinguishes fuel-combustion and fugitive-emission mitigation actions rather than treating the petroleum sector as a single undifferentiated emission source (Federal Republic of Nigeria, 2025).

In the transportation sector, no significant facility differences were observed for Sources, F (2, 398) = .074, p = .928; Types, F (2, 398) = .234, p = .792; HSI, F (2, 398) = .180, p = .836; or Mitigation, F (2, 398) = .868, p = .420. Thus, Transportation showed the greatest statistical consistency across facilities. This relative similarity may suggest common operational conditions within the surveyed transportation facilities. Nigeria's transport sector remains strongly dependent on fossil fuels, which can create common energy-use and decarbonization challenges across transport operators (Akujor et al., 2022). The present findings suggest that relatively standardized mitigation interventions may therefore be more feasible within the sampled transportation facilities than in sectors demonstrating greater facility heterogeneity.

Solid Waste showed no significant difference in GHG-source composite means, F (2, 395) = .509, p = .601. However, significant facility differences occurred for GHG Types, F (2, 395) = 11.375, p < .001, η^2 = .055; HSI, F (2, 395) = 3.536, p = .030, η^2 = .018; and Mitigation, F (2, 395) = 3.323, p = .037, η^2 = .017.

Tukey HSD showed that Facility A recorded significantly higher GHG-type scores than B and C. For health and safety impacts and mitigation opportunities, statistically significant differences were specifically observed between Facilities A and B.

The variation among Solid Waste facilities is consistent with evidence that different waste-management configurations can produce substantially different environmental outcomes and mitigation opportunities. Lawal et al. (2026), for example, demonstrated major differences between open dumping, burning, managed landfill, recycling, composting, and other waste-management scenarios. Consequently, Solid Waste appears to be the sector where facility-specific mitigation planning is most strongly supported by the present statistical evidence.

Taken together, the results provide three important findings. First, there is no universal correlation between identified GHG emission sources and mitigation opportunities across all facilities. Some statistically significant relationships occurred, especially within Solid Waste, while others were weak or absent. This finding is consistent with the wider mitigation literature, which demonstrates that decarbonization opportunities depend on activity type, available technology, infrastructure, and operating conditions rather than emission-source recognition alone (IPCC, 2022)

Second, facilities within the same sector are not necessarily statistically identical. Manufacturing facilities differed significantly in mitigation opportunities, Oil and Gas facilities differed in emission-source scores, and Solid Waste facilities differed on several constructs. Transportation was the principal exception, showing substantial similarity across all three facilities.

Third, the results demonstrate why mitigation planning should distinguish clearly between emission-generating activities, the gases produced by those activities, and the technological opportunities available to control them. For example, gas flaring is an activity that may produce CO_2 and other emissions; methane is a GHG that may arise from venting, fugitive releases, waste decomposition, or incomplete combustion. The two therefore cannot be compared as equivalent "emission sources."

This distinction is consistent with contemporary Nigerian mitigation priorities, which separately recognize fuel combustion, fugitive emissions, industrial activities, transportation, and waste management in national climate planning (Federal Republic of Nigeria, 2025). Most importantly, the current findings do not support the earlier assertion that there is a generally strong positive correlation between emission magnitude and reduction potential. The observed correlations range from non-significant to weak or moderate, and mitigation options differs among facilities.

Conclusion

The study examined relationships among identified GHG emission sources, GHG types, health and safety impacts, and mitigation opportunities within manufacturing, oil and gas, transportation, and solid waste facilities in Port Harcourt, Nigeria.

The findings demonstrated that GHG emissions and mitigation opportunities are sector- and facility-specific rather than universally consistent across the study population. The facilities exhibited broadly comparable categories of GHG generating activities, types and mitigation options, reflecting their common sectoral characteristics but despite this similarity in the categories identified, the strength of association in their quantitative emission/reduction profiles varied, indicating that the relative magnitude of emissions and reduction potential was not uniform across the sampled facilities. Manufacturing facilities displayed statistically similar source and GHG-type profiles but significant variation in mitigation opportunities, particularly between Facility C and Facilities A and B. Oil and Gas facilities demonstrated significant variation principally in identified emission sources. Transportation facilities were comparatively homogeneous, showing no significant facility-level differences across the four investigated constructs. Solid Waste demonstrated the greatest heterogeneity, with multiple significant positive correlations and significant facility-level variation in GHG types, health and safety impacts, and mitigation opportunities.

The findings therefore do not support the application of a single uniform GHG mitigation strategy across all facilities. Instead, interventions should reflect the particular emission-generating activities, GHG profiles, existing operational arrangements, and feasible mitigation technologies of individual facilities. This approach accords with international and Nigerian climate-mitigation frameworks that emphasize differentiated actions across energy, industry, transportation, oil and gas, and waste-management systems (Federal Republic of Nigeria, 2025; IPCC, 2022).

Quantifying actual emission reductions would require facility-specific activity data, emission factors, baseline GHG inventories, and technically feasible mitigation options.

Overall, the evidence supports facility-specific GHG accounting as an important complement to sector-wide climate mitigation planning. Such an approach can help direct mitigation efforts toward locally appropriate interventions while supporting Nigeria's broader transition toward lower-carbon industrial, transportation, energy, and waste-management systems.

References

Adiama, Y., Sawyerr, H., Opasola, A., Otto, E., & Akpan, N. (2025). Analysis of energy footprint of Port Harcourt metropolis, Nigeria: Implications for net-zero. *African Journal of Environment and Sustainable Development, 3*(2),93–104. https://doi.org/10.5281/zenodo.15799559 [openjourna.... ijaar.org]

Afinotan, U. A. (2022). How serious is Nigeria about climate change mitigation through gas flaring regulation in the Niger Delta? *Environmental Law Review, 24*(4), 288–304. https://doi.org/10.1177/14614529221137142 [journals.sagepub.com]

Akujor, C. E., Uzowuru, E. E., Abubakar, S. S., & Amakom, C. M. (2022). Decarbonization of the transport sector in Nigeria. *Environmental Health Insights, 16*, 1–8. https://doi.org/10.1177/11786302221125039 [journals.sagepub.com]

Eyetan, T. E., Ojoh, C. O., & Boyitie, P. O. (2025). Spatio-temporal variations of air pollutants in urban zones of Port Harcourt: Influence of climatic conditions and human activities. *International Journal of Research and Innovation in Social Science, 9*(7), 1615–1629. https://doi.org/10.47772/IJRISS.2025.907000134 [rsisintern...tional.org]

Federal Republic of Nigeria. (2023). *Nigeria: Facilitative sharing of views, second biennial update report*. United Nations Framework Convention on Climate Change.

Federal Republic of Nigeria. (2023). *Nigeria: Facilitative sharing of views, second biennial update report*. United Nations Framework Convention on Climate Change.https://unfccc.int/sites/default/files/resource/Nigeria%2014th%20workshop%20of%20FSV.pdf [unfccc.int]

Federal Republic of Nigeria. (2025). *Nigeria's third nationally determined contribution (NDC 3.0)*. United Nations Framework Convention on Climate Change. [unfccc.int]

Intergovernmental Panel on Climate Change. (2022). *Climate change 2022: Mitigation of climate change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press. https://doi.org/10.1017/9781009157926 [ipcc.ch], [ipcc.ch]

Intergovernmental Panel on Climate Change. (2023). *Climate change 2023: Synthesis report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. IPCC. https://doi.org/10.59327/IPCC/AR6-9789291691647 [ipcc.ch], [report.ipcc.ch]

Lawal, A. R., Akeredolu, F. A., Sonibare, J. A., & Adewale, A. J. (2026). IPCC-based scenario analysis of greenhouse gas mitigation and carbon credit potential in Lagos megacity waste. *The Open Chemical Engineering Journal, 20*. [openchemic...ournal.com]

Ogboeli, G. P., Dimkpa, G. C., Ogba, A. A., Ogbonda, P. N., Omodu, H. P.-E., Edim, E. E., & Adishome, C. A. (2025). A systematic review and meta-analysis of waste collection and transportation efficiency in urban waste management systems in Port Harcourt. *African Journal of Environment and Sustainable Development, 3*(2), 65–82. https://doi.org/10.5281/zenodo.15741499 [researchgate.net]

Raimi, M. O., Ezekwe, I. C., Agusomu, T. D., Enyinnaya, O., Amakama, N. J., & German, I. C. (2025). Enhancing methane emissions management in Nigeria's oil and gas sectors: A comprehensive policy and strategic framework. *Open Journal of Yangtze Oil and Gas, 10*(2), 31–62. https://doi.org/10.4236/ojogas.2025.102002 [scirp.org]